SMART CANTEEN

Online Food Pre-Ordering and
Canteen Management System

N.B.K.R. Institute of Science & Technology (NBKRIST)

Department of Computer Science and Engineering

M. Rishithanjali (24KB1A05Z9)
V. Aishwarya (24KB1A05MR)
K. Rohith (24KB1A05N4)
R. Jashwanth (24KB1A05FZ)
G. Reshma (24KB1A05K0)

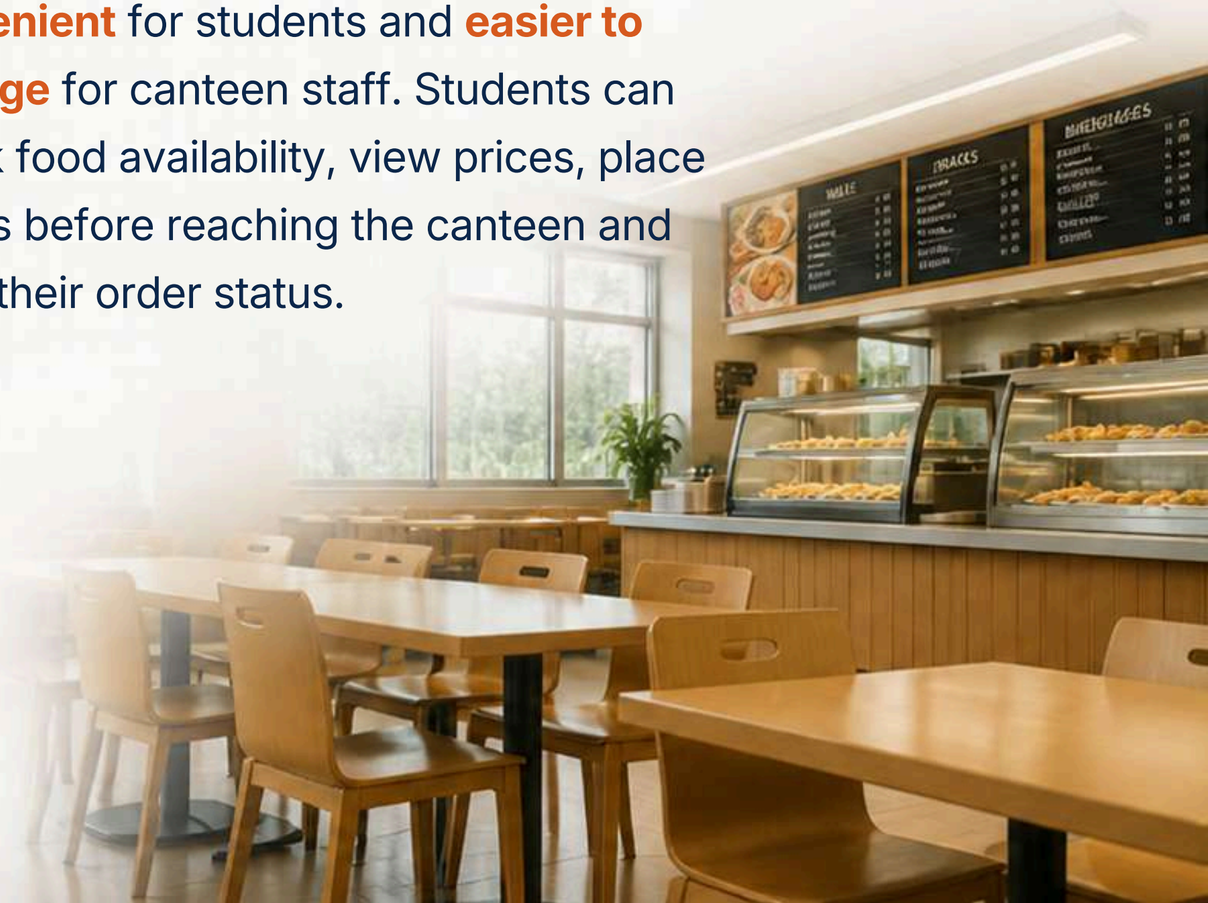


INDEX

1. Introduction
2. Problem Statement & Objectives
3. Modules & Requirements
4. Frontend / Similar Web
5. Conclusion & References
6. Questions / Suggestions

INTRODUCTION

Smart Canteen is an online food pre-ordering and canteen management system designed to make the college canteen experience more **convenient** for students and **easier to manage** for canteen staff. Students can check food availability, view prices, place orders before reaching the canteen and track their order status.



PROBLEM STATEMENT & OBJECTIVES

PROBLEM STATEMENT

In colleges, many students depend on the canteen for lunch and snacks, especially when they do not bring food from home. During break hours, the canteen becomes crowded and students have to wait in long queues. Since the break time is limited, this waiting can waste a significant amount of time and may cause students to be late for their next class. Another issue is that students usually do not know whether the food they want is available until they reach the canteen. If an item is unavailable, they may waste even more time deciding what to eat. On the canteen side, managing a large number of orders manually can lead to confusion, delays, and incorrect orders. Therefore, we propose a Smart Canteen – Online Food Pre-Ordering and Management System, where students can check food availability, view prices, pre-order their food, and track their order status. The system also helps canteen staff manage orders and food availability efficiently, reducing queues, waiting time, and order-related confusion.

OBJECTIVES

- Reduce waiting time and queues
- Allow students to check food availability
- Display food items and prices
- Enable food pre-ordering
- Provide order-status tracking
- Help canteen staff manage orders efficiently
- Reduce manual confusion and incorrect orders

KEY ISSUES



Long Queues



Waiting Time



Food Availability



Manual Order Confusion

MODULES & REQUIREMENTS

STUDENT MODULE

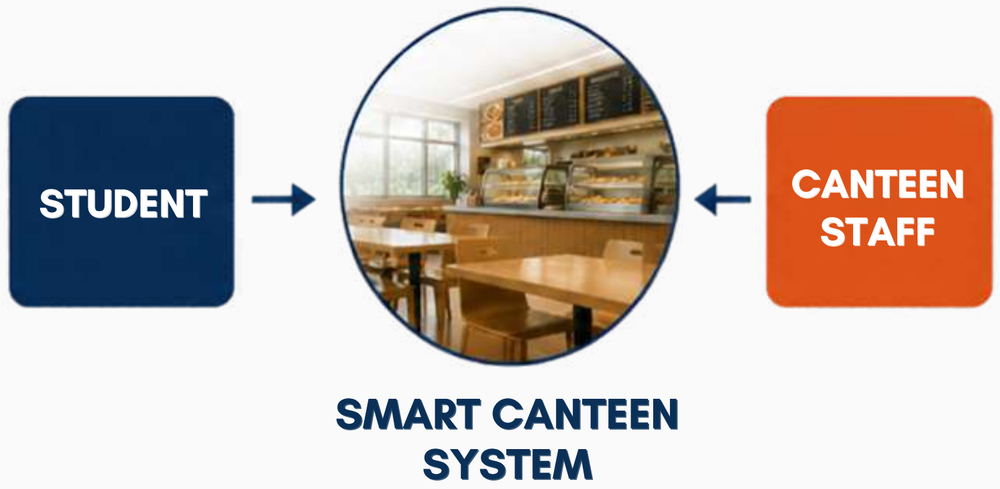
- Login/access system
- View food menu
- Check availability
- View prices
- Select food items
- Place pre-order
- Track order status

CANTEEN STAFF/ADMIN MODULE

- Manage food items
- Update food availability
- View incoming orders
- Manage orders
- Update order status
- View completed orders

FUNCTIONAL REQUIREMENTS

- User access/login
- Menu management
- Food availability management
- Order placement
- Order status update
- Order tracking

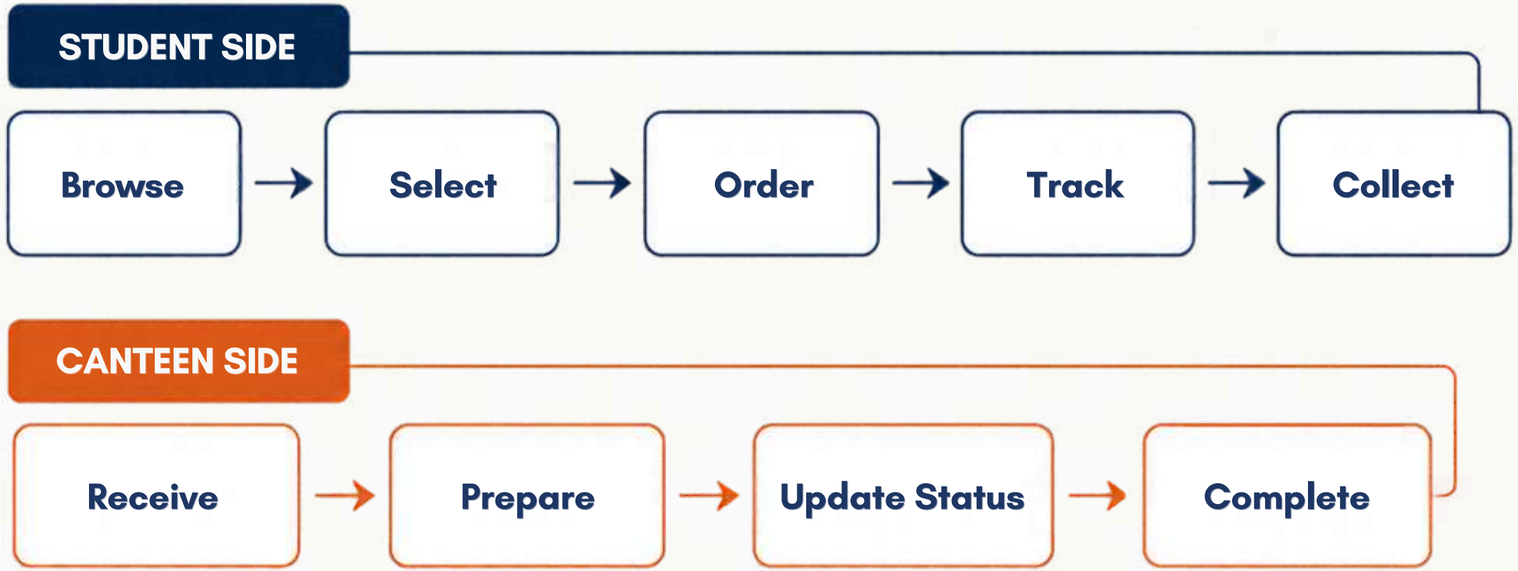


BASIC SYSTEM REQUIREMENTS

- Frontend interface
- Backend/application logic
- Database
- Internet network access

HOW THE SMART CANTEEN WORKS

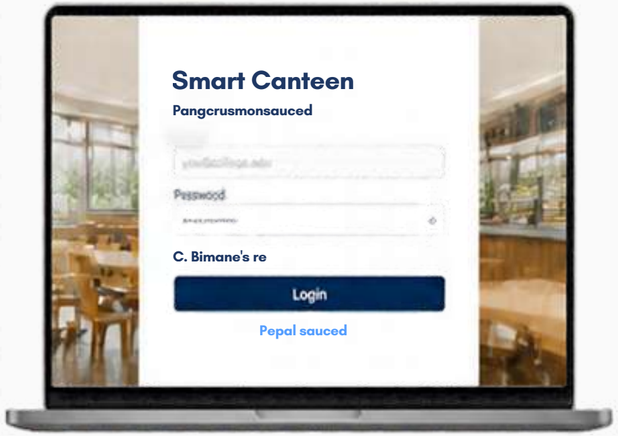
1. Student opens system
2. Views available food
3. Checks prices and availability
4. Selects food
5. Places pre-order
6. Canteen receives order
7. Canteen prepares food
8. Order status is updated
9. Student collects order



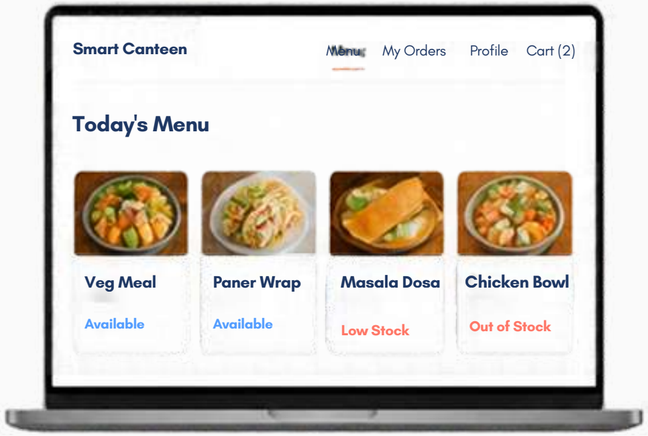


FRONTEND / WEB INTERFACE

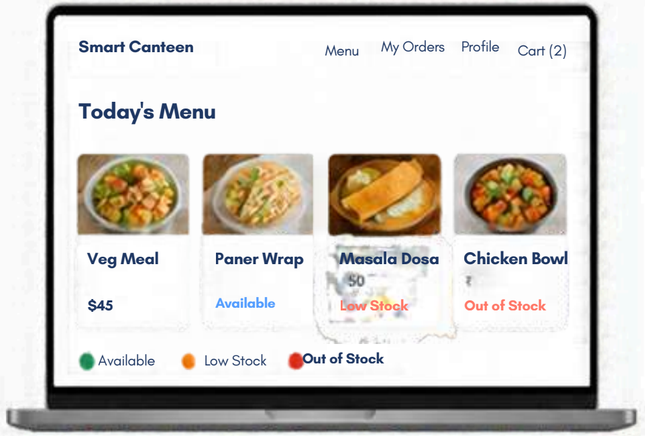
Login Page



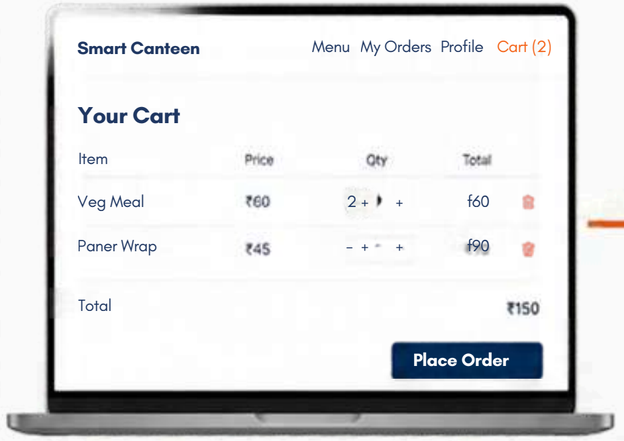
Home / Menu Page



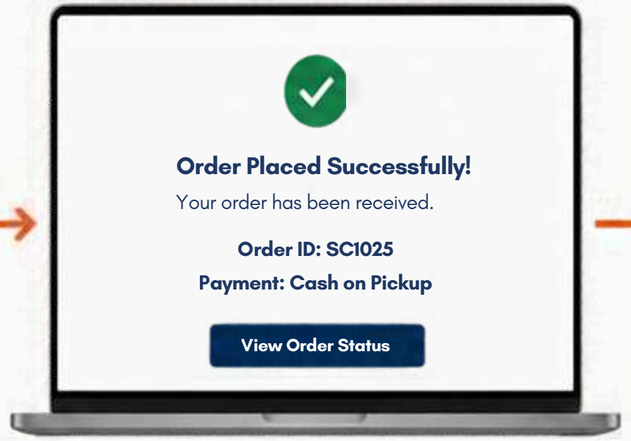
Food Availability Indicators



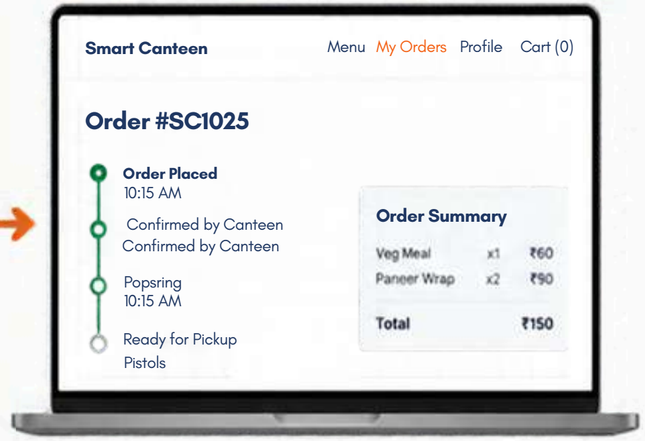
Cart / Selected Items



Order Placement Confirmation



Order Status Tracking



FRONTEND FEATURES

- Simple navigation
- Clear food menu
- Availability indication
- Easy ordering
- Order-status display
- User-friendly interface

SYSTEM REQUIREMENTS

FRONTEND

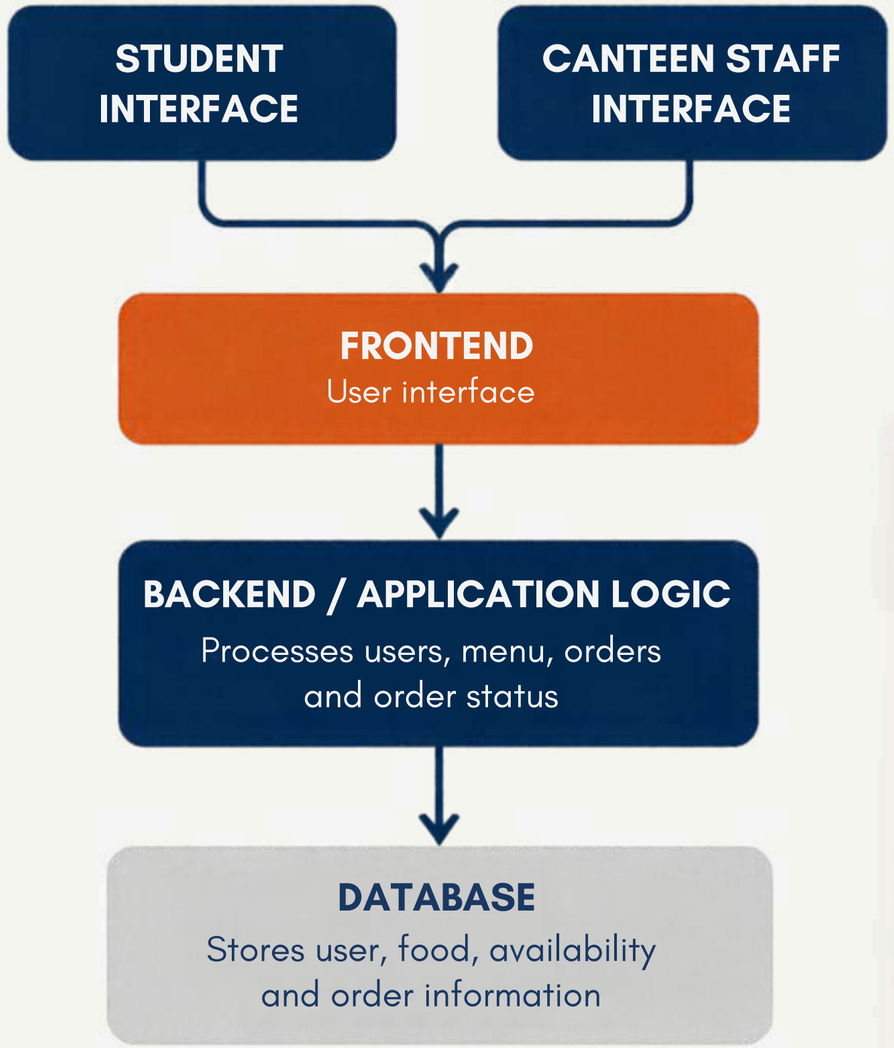
User interface for students and canteen staff

BACKEND

Handles users, menu, orders, and order status

DATABASE

Stores user, food, availability and order information





BENEFITS OF SMART CANTEEN

FOR STUDENTS

- Less waiting in queues
- Saves break time
- Easy menu access
- Know food availability before ordering
- Convenient pre-ordering
- Order-status tracking

FOR CANTEEN STAFF

- Better order organization
- Reduced manual confusion
- Easier food availability updates
- Better handling during busy hours
- Clear order-status management
- Reduced chances of incorrect orders



EXPECTED OUTCOME

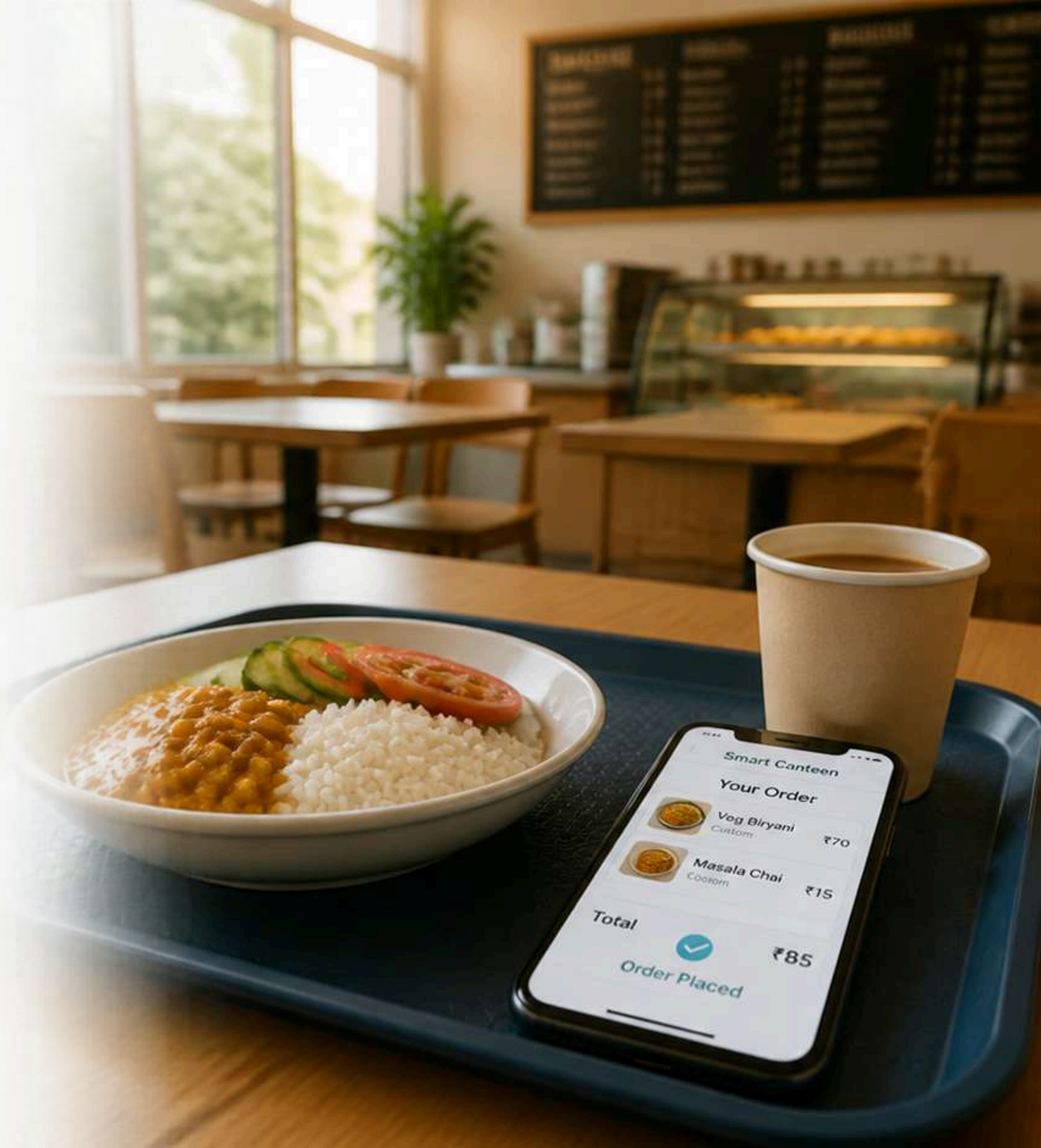
A faster, simpler and more organized college canteen ordering experience.

CONCLUSION

Smart Canteen provides a simple digital solution to common problems faced by students and canteen staff. By allowing students to check availability and **pre-order food,** the system can reduce unnecessary waiting and make order handling more organized.

REFERENCES

HTML, CSS and JavaScript documentation
Database documentation
Web development learning resources
User interface design references



QUESTIONS / SUGGESTIONS?

THANK YOU

SMART CANTEEN – Online Food Pre-Ordering and Canteen Management System

